SAMARTH

CUET PG 2026

Computer Science | **Lecture - 19**

Unit : Operating System
Topic : Previous Year Questions + Doubt Class

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Recap *of previous lecture*

1. Unix and Windows



Topics *to be covered*

1. Previous Year Questions + Doubt Class





Topic : Previous Year Questions + Doubt Class

P4 445 + 311 = 756 + 322 = 1078 <P2, P4, P1, P3>

<P4, P2, P1, P3>

Q:

Processes	Max R1 R2 R3	Allocated R1 R2 R3	Need
P1	8 6 4	1 2 1	7 4 3
P2	4 3 3	3 1 1	1 2 2
P3	10 1 3	4 1 3	6 0 0
P4	3 3 3	3 2 2	0 1 1
		11 6 7	

Available R1 R2 R3: 4 4 5

+322 = 767 + 311 = 1078

What is correct?

(A) 1, 2, 4

(B) 1, 2, 3

(C) 2, 3, 4

(D) 1, 2, 3, 4

Consider 4 safe sequences:

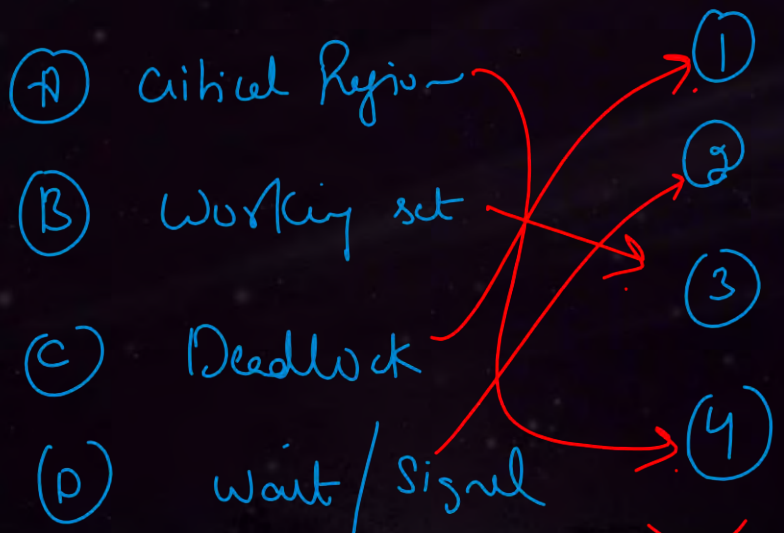
(1) P2 P4 P1 P3

(2) P4 P2 P1 P3

(3) P4 P2 P3 P1

(4) P1 P4 P2 P3

Q. Match the following.



(1) A-4, B-3, C-1, D-2

(2) A-4, B-3, C-2, D-1

(3) A-3, B-4, C-2, D-1

(4) A-3, B-4, C-1, D-2.

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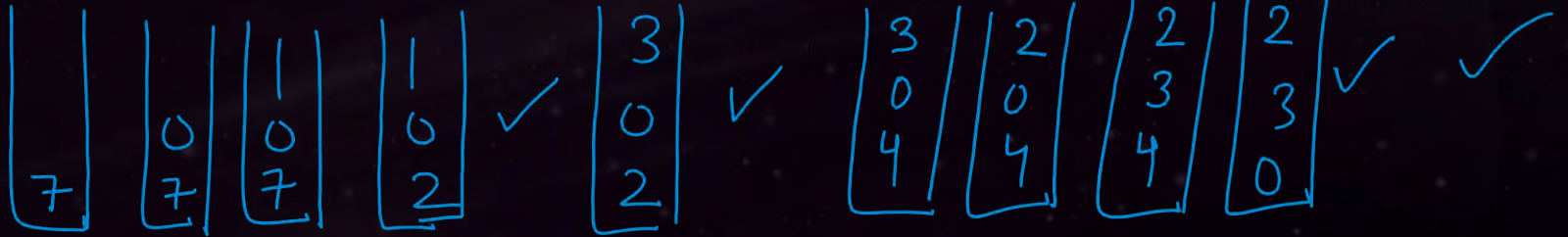
LRU

Calculate no of page faults?

3 page frames.

Q: Assume page Reference string as

7 0 1 2 0 3 0 4 2 3 0 3 2



(1) 10 (2) 8 ~~(3)~~ 9 (4) 11

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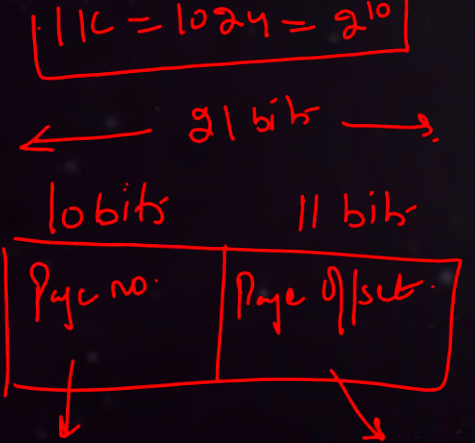
Q:// Consider a system with 1K pages and 512 frames & each page is of size 2KB. How many bits are required to rep. Virtual add space memory.

(1) 20 bits
(2) 21 bits
(3) 11 bits
(4) 16 bits

2^{10}

$2^1 \times 2^{10} = 2^{11}$

1K = 1024 = 2^{10}



Q. Which of the following scheduler/ Schedulers is also called as CPU scheduler?

(A) Short Term scheduler — CPU scheduler.
(B) Long " " — Job scheduler.
(C) Medium " " — Process swapping scheduler.
(D) ~~Asymmetric " .~~

(A) A, B, D
(B) A, B, C
(C) A, B, C, D
(D) A only.

Q. How does no. of page frames affect no of page faults in FIFO page Rep?

(A) ↑ no of frames always ↓ no of page faults. ✗

(B) ↑ " " " may ↑ or ↓ no of page faults depending upon memory access pattern. ✓

(C) ↑ no of frames alway ↑ no of page faults. ✗

(D) ↑ " " " has no effect on no of page faults. ✗

Belady anomaly ✓

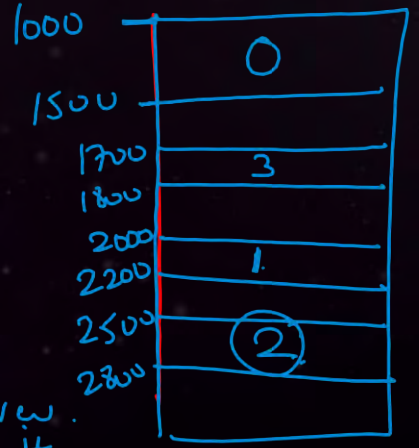
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Q. On a system using Segmentation, full is segment table.

Segment	Limit	Base
0	500	1000
1	200	2000
2	300	2500
3	100	1700

Add given limit + Base add.

212 + 2500 = 2712



given limit.

What is the physical add for logical add (2), (212)?

(A) 2712 (B) 512 (C) 2212 (D) 2800

limit ≤ 300

212 ≤ 300

Yes.

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Q How many child processes will be created for total fork() system call?

n - no of times Fork is written

```
Fork() ;
Fork() ;
Fork() ;
Fork() ;
```

$= 2^n - 1$

$= 2^4 - 1$

$= 16 - 1$

$= 15$

① 4 ② 16 ③ 15 ④ 3

No of times "Hi" is printed $\Rightarrow 2^n$

Thank You